

Recreating a Functioning Forest Soil in Reclaimed Oil Sands in Northern Alberta: An Approach for Measuring Success in Ecological Restoration

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During oil-sands mining all vegetation, soil, overburden, and oil sand is removed, leaving pits several kilometers wide and up to 100 m deep. These pits are reclaimed through a variety of treatments using subsoil or a mixed peat-mineral soil cap. Using nonmetric multidimensional scaling and cluster analysis of measurements of ecosystem function, reclamation treatments of several age classes were compared with a range of natural forest ecotypes to discover which treatments had created ecosystems similar to natural forest ecotypes and at what age this occurred. Ecosystem function was estimated from bioavailable nutrients, plant community composition, litter decomposition rate, and development of a surface organic layer. On the reclamation treatments, availability of nitrate, calcium, magnesium, and sulfur were generally higher than in the natural forest ecotypes, while ammonium, P, K, and Mn were generally lower. Reclamation treatments tended to have more bare ground, grasses, and forbs but less moss, lichen, shrubs, trees, or woody debris than natural forests. Rates of litter decomposition were lower on all reclamation treatments. Development of an organic layer appeared to be facilitated by the presence of shrubs. With repeated applications of fertilizers, measured variables for the peat-mineral amendments fell within the range of natural variability at about 20 yr. An intermediate subsoil layer reduced the need for fertilizer and conditions resembling natural forests were reached about 15 yr after a single fertilizer application. Treatments over tailings sand receiving only one application of fertilizer appeared to be on a different trajectory to a novel ecosystem.

THERE are an estimated 1.6 trillion barrels of oil within the oil sands located in the Peace River, Cold Lake, and Athabasca River regions of northern Alberta, of which 311 billion barrels in the Athabasca region are believed to be recoverable with current surface mining technology (Government of Alberta, 2007: www.albertacanada.com/industries/924.html). During mining, all vegetation and soil cover is removed and about 15 to 50 m of overburden is stripped away to reveal the oil sand, which is then itself stripped away and transported to the extraction plant. This creates open pits several kilometers wide and up to 100 m deep into which unwanted mine residues may be tipped. The oil companies operating in the region are obliged to reclaim their operational land as set out within the Land Surface Conservation and Reclamation Act 1973 and the Environmental Protection and Enhancement Act 1992 (Government of Alberta, 1999). Provincial guidelines for reclamation require the oil companies to establish commercial forest within the natural range of ecotypes found in the Central Mixed Wood Sub-Region of the Boreal Forest and, secondarily, wildlife habitat capabilities similar to predisturbance conditions (Government of Alberta, 1999). The reclamation of the Athabasca oil sands mines is a challenging soil-forming and landscaping exercise on a massive scale.

The reclamation treatments used by the oil companies use the mine residues in combination with overburden, stock-piled soil and peat, to create soil-like materials capable of supporting stable plant and microbial communities. Over the last 30 yr seven main prescriptions have been used; these include peat-mineral mix or subsoil or both over tailings sands, overburden, Clearwater shale or lean oil sands (Table S1).

During removal, the shale overburden becomes fragmented and increases in volume, allowing water to percolate through it, leading to weathering, slumping, and erosion (Stolte et al. 2000). For these reasons all saline overburden is capped with 80 cm of nonsaline overburden or till. The properties of nonsaline overburden vary; those salvaged near the surface (up to 1 m below the soil surface)

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Abbreviations: FH, fermented/humified surface organic matter; MRRP, multi-response permutation procedures; NMS, nonmetric multidimensional scaling; P-M, peat-mineral; PRS, plant root simulator.

have variable clay content and pH between 4.9 and 6.6. The overburden 1 to 2 m below the soil surface has higher pH (8) and higher base-cation content (Danielson et al., 1983). Before being mined, the sand is compressed due mainly to the effect of historic glaciation, but after mining and processing the sand occupies a greater volume than before. Thus even after the removal of the oil component, the mining residues occupy a greater space than previously, thus reclaimed sites other than wetlands tend to be raised areas rather than depressions.

The organic component of reconstructed soils is predominantly peat, which is available in large amounts in the mining footprint. Peat is usually added as a 20 to 50 cm capping to the reconstructed soils to increase their organic C and nutrient content, and improve their water-holding capacity. The peat-mineral soil is coarsely mixed during the salvage operation during which peat is over-stripped to a maximum depth of 3 m and so includes 25 to 50% by volume of mineral materials (Fung and Macyk, 2000). Where lean oil sands (unprocessed oil sands with <6% oil by mass) and saline overburden, rather than tailings sand, is being reclaimed, an intermediate capping layer of up to 80 cm of nonsaline subsoil material is used to bury the material and protect it from water and plant root ingress (Oil Sands Vegetation Reclamation Committee, 1998; Stolte et al., 2000; C. Qualizza, personal communication, 2005). There is as yet no standard prescription or “recipe” for reclamation—the oil companies adopt different methods of reclamation with different materials, different cap depths and, experimentally, replace peat with salvaged forest floor from upland areas (McMillan et al., 2007; C. Welham personal communication, 2005).

During the first growing season after capping, barley (*Hordeum vulgare* L.) is sown to provide quick vegetation cover and erosion control. Barley is a poor competitor and is readily invaded by local flora within the first few years (Hardy BBT Ltd, 1990). Natural plant recolonization from adjacent reclaimed sites has been successful for grasses and forbs, but shrub and tree establishment has not been significant and tree planting is deemed necessary (Hardy BBT Ltd, 1990; W. Tedder, personal communication, 2005). Nursery-grown white spruce [*Picea glauca* (Moench) Voss] and aspen (*Populus tremuloides* Michx. Lamb) are usually planted on sites being reclaimed for forestry, usually at the next available planting season. Some companies practice fertilization at tree planting and at regular intervals postplanting; others do not fertilize.

The overall aim of this research is to determine (i) if the reclamation practices are creating ecosystems which appear to be on a predictable path to become functioning forest ecosystems similar (i.e., within the natural range of variability) to those naturally present in the region (Fig. 1), (ii) which type(s) of amendments are most successful, and (iii) by what age this occurs. An estimate of the natural range of variability for key response variables in reference (or “target”) natural forest ecotypes were compared with those of a number of oil sands reclamation treatments across a range of ages. The response variables measured included two of the three major attributes of reclamation identified by Ruiz-Jaen and Aide (2005): vegetation structure and ecological processes. Vegetation was assessed through ground cover estimates, and ecological processes were

assessed by measuring nutrient bioavailability, litter decomposition rates, and development of a surface organic layer.

Materials and Methods

Study Area

The study area is within the Wood Buffalo region of northern Alberta, Canada. The study area covers an area of about 135 km north-south and 50 km east-west, centered approximately 35 km north of Fort McMurray (57°00' N, 111°28' W; 369 m above sea level). Daily average temperatures range from −18.8°C in January to +16.8°C in July, and only 2 mo of the year are deemed frost-free. Average annual precipitation is 455 mm of which 342 mm falls as rain (Environment Canada Climate Normals 1971–2000; www.climate-weatheroffice.ec.gc.ca).

The oil sands are the delta deposits of an ancient tropical sea, subsequently compressed by extensive glacier formations and invaded by oil migrating from elsewhere. The local Clearwater shale formation north of Fort McMurray lies over the oil-rich sand deposits and is saline (Mossop, 1980; Marshall, 1982); these consolidated deposits have strongly sodic or saline pore fluids where electrical conductivity exceeds 6 dS/m (Stolte et al., 2000; Purdy et al., 2005) and contain highly dispersive sodium-clays that are stable only while confined in situ. The shales are physically stable while at an angle of repose not exceeding 7° or 12%, resulting in a gently undulating landscape. Glacial till from the Pleistocene Epoch-era forms a clay-rich mineral soil layer over this except in some places where fluvial sand persists (Stolte et al., 2000), supporting a mosaic of communities from bogs and rich or poor fens to dry pine forests.

Natural soils within the oil sands are classified broadly as Brunisols, Luvisols, Gleysols, or Organic, this following a moisture gradient from xeric to subhydric and from aerobic to increasingly anaerobic and reducing conditions (Turchenek and Lindsay, 1982; Oil Sands Vegetation Reclamation Committee, 1998). This gradient in soil development is caused by impeded water flow due to increasing clay content in the mineral subsoils, from glacio-fluvial sands to glacio-lacustrine muds to fine-clay glacial till. Gray Luvisols have developed on medium to fine-textured lacustrine deposits and till, and Dystric Brunisols have developed on coarser parent material such as glaciofluvial outwash and eolian sands. Organic soils developed on poorly drained lowlands overlaying glacial deposits (Natural Regions Committee, 2006) provide the bulk of the raw peat that is salvaged for use in reclamation.

The dominant tree species in the Central Mixed Wood Sub-Region of the Boreal Forest are aspen, poplar (*Populus balsamifera* L.), white spruce, black spruce [*P. mariana* (Mill.) Britton, Sterns and Poggenb.], jack pine (*Pinus banksiana* Lamb), balsam fir [*Abies balsamea* (L.) P. Mill], and tamarack [*Larix laricina* (Du Roi) Koch] (Rowe, 1972). The forest has been classified into several ecotypes according to landscape, drainage, and soils, with different plant communities developing with different proportions of tree species including jack pine, aspen, and white spruce (Beckingham and Archibald, 1996). Ecotypes with the potential end-use of commercial forestry are denoted by the predominant

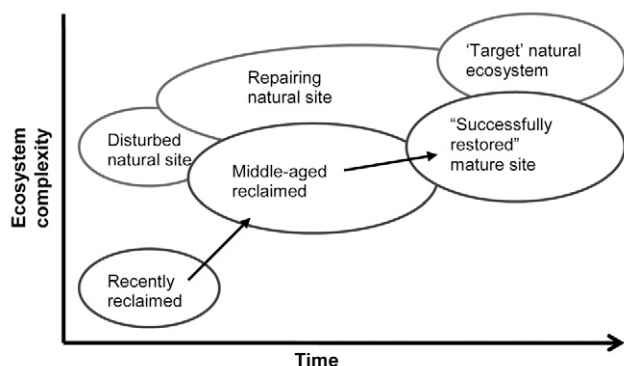


Fig. 1. A model for ecosystem restoration; arrows indicate one possible restoration trajectory, or predictable path, to a successful conclusion.

indicator species- blueberry (b) (*Vaccinium* spp.), Labrador tea-mesic (c), low bush cranberry (d) (*V. macrocarpon* Ait), dogwood (e) (*Cornus* spp.), and horsetail (f) (*Equisetum* spp.) (Oil Sands Vegetation Reclamation Committee, 1998).

Site Descriptions

Twenty-nine reclaimed sites were selected, representing a range of age (3–34 yr) and treatments practised by the three major oil companies operating in the Athabasca region around Fort McMurray, Syncrude Canada Ltd, Albion Sands Energy Inc., and Suncor Energy Inc. (Table 1). All but one of these are long-term monitoring plots established between 2000 and 2006; the remaining plot (WA5, a recolonized peat waste area in the Suncor facility) was established for this study.

Eighteen natural sites were selected from the long-term monitoring plots; these ranged from xeric/nutrient-poor sites (ecotype a1) to mesic/nutrient medium-to-rich sites (ecotype d1) (Table S2). Three replicates of each of six natural ecotypes were drawn from ecotypes a1, b1, b3, d1, d2, and d3. Ecotypes c, e, and f were not included because ecotype c is not found locally, and ecotypes e and f represent very wet forests that are not likely to be created by the reclamation treatments. One plot representing a b3 ecotype was destroyed mid-study by the expansion of a mine. All of the reclaimed and several of the natural forest sites are within the mine lease areas of Syncrude, Suncor, and Albion Sands; the remainder are natural forest sites located on public land outside of the leased mining areas. Each long-term monitoring plot measures 10 by 40 m; all measurements and samples were taken around the plot perimeter (within 2 m of the plot) to preserve the integrity of the environment within the plot.

Sampling

Unless stated otherwise, “soil” is the term used to refer to the upper mineral component on a natural site or the original capping layer, whether a peat-mineral (P-M) mixture or subsoil, of a reclamation treatment. Soil is, therefore, the mineral-rich component found closest to the surface, capable of supporting vegetation growth and developing an organic layer over its surface. By this terminology the capping layer is analogous with the upper mineral layer in a natural soil.

Soils were sampled soon after spring thaw in June 2005, with the exception of Albion and FF plots which were sampled in June 2006. Sampling was semi-systematic; a composite sample was made from 10 soil cores (7-cm diam.) taken from four locations down each long side, and one at each short side of the plot. As each core was extracted it was laid out on a plastic bag and the litter from the top of the core was hand-picked into a plastic bag. The depth of the surface organic layer (hereafter referred to as FH) was measured before it was put into a separate plastic bag. The F and H layers were harvested together because they could not be distinguished in the field with confidence. On younger reclaimed sites and most subsoil-capped treatments, the organic layer occurred patchily as a thin black film without an obvious F layer. On some reclaimed sites close examination was necessary to determine where the thin FH ended and the original P-M mixture began, the FH generally being darker than the P-M. Any live vegetation was removed at this stage. Finally a handful of soil from the upper 10 cm was taken. Samples were composited $\times 10$, well-mixed by hand, and stored in resealable plastic bags on ice until the end of the day. Litter and FH were then weighed wet. All bulk samples were frozen (-18°C) at the Syncrude facility until transported to the laboratory for subsequent drying and analysis.

Moisture Content, pH, and Carbon/Nitrogen Ratio

On receipt at the lab, the frozen samples of soil, FH, and litter were defrosted overnight and a weighed subsample was placed in a drying oven (105°C) for 72 h and reweighed to obtain gravimetric moisture content and total dry mass. A second subsample was dried at about 50°C for several days before being analyzed for total C and N by automated dry combustion using a Carlo-Erba elemental analyzer with the remainder analyzed for pH in water [pH(w)]. All samples of the FH and soil were sieved (<4 mm) to remove stones and roots before analysis.

Development of the Surface Organic Layer

The mass of the FH layer in each plot was estimated from the original wet FH sample mass adjusted by accounting for gravimetric moisture to obtain dry mass per m^2 (the original sample being a composite of 10 7-cm cores, total area harvested 0.038 m^2). The dry mass (i.e., $\text{g}/0.038 \text{ m}^2$) and the mean ($n = 10$) FH depth were transformed to natural log $\ln(x+1)$, to achieve a normal distribution. The (transformed) FH dry mass and depth, pH, total N, total C, and moisture content were regressed against site age to find a model for FH development. The age by which FH development would fall within the natural range of variability was estimated from the model.

Litter Decomposition

Mass loss of aspen leaf litter was measured to compare litter decomposition rates across the range of reclamation treatments and natural ecotypes. In September 2005, fresh-fallen or recently senesced litter samples of naturally-growing aspen were collected from a forest within the study area. The litter was stored in paper bags and allowed to dry indoors for at least 48 h before use. A subsample was stored dry for later analysis of total C and N concentrations.

Table 1. Reclamation treatments (L. Leskiw, personal communication, 2005; C. Qualizza, personal communication, 2005, 2006; W. Tedder, personal communication, 2007).

Depth, cm	A	B	E	F	H	I	M	Fireweed fibric	Waste Area 5	Albian sands
10	Peat-Mineral	Subsoil	Peat-Mineral	Subsoil	Peat-Mineral	Peat-Mineral	Mesic peat- Mineral	Fibric peat- Mineral	Peat	Peat-Mineral
20										
30										
40	Subsoil		Subsoil							
50										
60						Overburden (Shales)	Subsoil	Subsoil		
70					Tailings sand					Tailings sand
80	Tailings sand		O'burden (Shales)	O'burden (Shales)						
90		Tailings sand								
100							Clearwater shales	Clearwater shales		Lean oil sand
....										
Comp.		Syncrude			Suncor		Syncrude		Suncor	Albian
Plots (Age)	36(11) 37(24) 40(14) 87(5) 88(4) 89(5)	3(14) 38(19) 39(9) 42(11) 46(10)	7(12) 14(10) 43(22)	12(14)	1(14) 16(9) 17(13) 24(34) 30(34)	25(21) 28(22) 32(21) 34(24) 75(3)	86(6)	FF(4)	WA5 (29)	Albian(3)
Fertilizer (N: P: K)		1 × 250–350kg/ha† (35:46:44)			1 × 300 kg/ha† (71:33:20) 4 × 250kg/ha/yr† (315:70:41)		250– 350 kg/ha† (35:46:44)	336kg/ha (10: 30: 15)	None	None

† Lanoue (2003).

Litter bags were made from 10 by 10 cm squares of nylon mesh fabric (mesh size 0.8 by 0.125 mm) sewn on three sides with polyester thread. Litter samples of approximately 0.50 g were placed inside a bag with a numbered metal tag and the open end was sealed using a hot-glue gun and general-purpose glue sticks.

In September 2005 seven bags per site were placed on top of the soil on a single replicate of each reclamation treatment and natural forest ecotype (except plots Albian and FF which were not added to the study until 2006). All bags were retrieved after 365 (± 4) d, air-dried, and the contents were removed and weighed. Only three bags were usable from treatment H as the bags had become buried by tailings sands which had concreted to the decomposing aspen litter. The sand in these bags was carefully removed with a fine sable brush. There was no visible contamination in any of the other bags.

In situ Bioavailable Nutrients

Plant root simulator (PRS) probes (Western Ag Innovations Inc., Saskatoon, SK, Canada) were used to measure bioavailability of nutrients within the soil. The probes consist of cation- and anion-exchange resin membranes encased in a plastic holding device, which are inserted into soil to measure nutrient supply in situ with minimal disturbance (Qian and Schoenau, 2002). The PRS probes were inserted in four random locations on each plot, with pairs of probes (one cation, one anion) at each location, a total of eight probes per plot. The probes were first installed in June 2006 and retrieved after 44 (± 6) d, whereupon fresh PRS probes were installed and incubated for a further 51 (± 6) d. Probes were washed with deionized water, bulked according to treatment and sent to Western Ag Innovations for analysis. Anions and cations from the probes were eluted for 1 h using 0.5 N HCl/2 mol L⁻¹ KCl. The eluate was analyzed for levels of ammonium (NH₄⁺) and nitrate (NO₃⁻) using automated colorimetry. Inductively-coupled plasma (ICP) spectropho-

tometry/atomic absorption spectrometry (AAS)/flame emission spectrometry (FES) was used to measure levels of P, K, S, Ca, Mg, Al, Fe, Mn, Cu, Zn, B, and Pb in the 0.5 N HCl/2 mol L⁻¹ KCl eluate. The totals for the two burial periods were combined and converted to micromoles per 10 cm² per total burial period. The PRS probe nutrient capture is relative among sites, it does not equate to total soil nutrient flux over the term of incubation and there is no linear relationship of capture rate vs. time.

Plant Community

During July and early August 2006 groundcover was surveyed on all plots except plot 64 which had been destroyed. Ten 1 by 1 m quadrats were laid systematically around each plot starting at one corner and at 10-m intervals along each long side and at the 5-m midpoint of the ends. Within each quadrat a visual estimate of all layers of groundcover <2 m above ground level was scored as percent cover (one 10 by 10 cm square = 1%), classified as: pine, spruce, broadleaf, woody shrubs, forbs, grasses, mosses, lichens, woody debris (>5 cm long or >1 cm diam.) and bare ground. Aerial cover above 2 m was estimated visually and the percent cover was added to the score for that vegetation type.

Statistical Interpretation

The experimental design was based on the availability of reclamation treatments; there are numerous confounding factors including uneven numbers of replicates (sometimes only one per treatment), age since reclamation, age at time of planting, planting species, proximity to other reclaimed sites or disturbed areas (related to natural colonization potential), fertilization treatment, peat type, and prior stock-piling (if recorded), altitude, and aspect. This created problems with using classical statistical methods to identify treatment effects. Where possible, data were analyzed using SAS 9.1 (SAS Institute Inc., Cary, NC) to identify models and treatment effects using clas-

sical statistical methods. PC-Ord 5.0 (MJM Software Design, Gleneden Beach, OR) had specific relevance to biological community data (McCune and Grace, 2002) and was used to apply nonparametric analyses where appropriate.

Nonmetric Multidimensional Scaling

An ordination technique known as nonmetric multidimensional scaling (NMS), was chosen as it is able to manipulate a large range of variables, normality and linear relationships are not a prerequisite for analysis (McCune and Grace, 2002), and it has been used to evaluate community-environment relationships in ecosystems (Laughlin and Abella, 2007). It produces an image of the plots in an ordination space that provides a visual aid to interpreting similarity among entities, those more similar being clustered closer together. Groundcover data from quadrat surveys were transformed from raw percent cover to $2/\pi$ arcsine square-root(x) as this has the effect of equalizing the class limits for very high and very low proportion values while maintaining a broad spread of classes for intermediate values (McCune and Grace, 2002).

The NMS ordinations were undertaken using PC Ord v.5.0 (McCune and Grace 2002, and references therein). Two data matrices were used, one containing groundcover data and the second containing chemical data for the soils (pH(w), C/N ratio, moisture content, bioavailable nutrients) as well as categorical information (treatment or age class, the latter by 5-yr increments, except natural ecotypes which were arbitrarily assigned to a top age class). Mass loss by decomposition was not used in this analysis since only one replicate per treatment had received the bags, so creating an imbalance in the spread of data. An NMS ordination was performed on the groundcover data, and the chemical variables in the second matrix were subsequently correlated and overlaid on the NMS axes. This suited our objective of assessing the restoration treatments primarily on the degree of vegetation recovery followed by an independent evaluation of relationships with potential causal factors.

The distance measure used was Sorensen (Bray-Curtis). The “auto-pilot” and “medium thoroughness” options were selected. This undertook a series of repeated computations (iterations) to adjust the position of entities (plots) in ordination space. It began with a four-dimensional solution and reduced to one dimension for each of 50 runs with real data starting with random coordinates, then 50 runs with shuffled data (that is, shuffled within columns—each column being one variable). This shuffling forms the basis for a Monte Carlo test of significance of each dimensionality (McCune and Grace, 2002). The best solution was the number (n) of dimensions beyond which additional dimensions do little to reduce stress. Stress is the departure from monotonicity in the plot of distance in the original n -dimensional space vs. distance in the ordination space, thus the closer data points lie to a monotonic line, the lower is the stress (McCune and Grace, 2002). A final stress of <20 was deemed satisfactory for ecological community data (McCune and Grace, 2002). A plot of stress vs. iteration (and the maximum number of iterations was 200) showed if and when the stress declined to a constant, and at that point was deemed to be stable, or if it fluctuated (in which case, more iterations required), with a final instability value automatically calculated from the standard deviation

in stress over the preceding x iterations (McCune and Grace, 2002); an instability criterion of 0.00001 was set as a “target” by the auto-pilot mode. The auto-pilot then ran a final, best solution (low stress, low instability value) using starting coordinates taken from an automatically-saved file for a previous run that had produced the better solution in n dimensions.

After the first NMS ordination, stress levels were too high (>35, but stable), so pine and spruce were additively combined to create a new variable of “conifer” that was transformed as before. The NMS ordination was repeated with this new variable in the groundcover matrix.

The resulting three-dimensional image was rotated on screen until a subjective decision was made to select a “best view” showing a relatively clear separation between groups. The categories within the viewing options were then changed from “treatment” to “age class” and the new image was viewed. Correlation coefficients (r) expressing linear relationships of each variable with each axis were revealed by selecting the Pearson Correlation option that, in PC-Ord 5.0, becomes available by viewing a two-dimensional, rather than three-dimensional, image; the square (r^2) expresses the proportion of variation on that axis that is “explained” by the variable in question (McCune and Grace, 2002).

Multi-Response Permutation Procedures

Multi-response permutation procedures (MRPP) were undertaken in PC Ord 5.0. The MRPP is a nonparametric analysis of within-group and between-group differences, to test for significant differences among two or more groups. The results provide a measure of heterogeneity or homogeneity while giving a P value to distinguish between pairs of groups where $n > 1$ (McCune and Grace, 2002). The MRPPs were undertaken first by treatment vs. natural ecotypes (grouped together) and, next, by treatment (where $n > 1$) vs. each ecotype from a1 to d3. The distance measure chosen was Sorensen (Bray Curtis). A test statistic, the chance-corrected within-group agreement was examined—this indicates whether entities within groups are identical to one another (value = 1.0) or as different as would be expected by chance (value = 0) or more different than would be expected by chance (value < 0), with typical community ecology values being <0.30 (McCune and Grace, 2002).

Cluster Analysis

To further examine relationships among the reclamation treatments and the natural ecotypes a cluster analysis using PC Ord v.5.0 was performed on the ground-cover (biological) data. In this method, cluster groups of plots were identified from a dendrogram and inspected for which reclaimed sites were clustered with natural ecotypes, and which were not. As in the NMS, the environmental (chemical) data were used in a subsequent step to help interpret the clustering results.

Results

Moisture Content, pH and Carbon/Nitrogen Ratios

Soil moisture in the reclamation prescriptions increased in order: F < H < Albion < B < I < WA5 < M < A < E < FF (Fig. S1). This

rank order will be followed in each of the graphs. Generally, natural ecotypes followed the expected trend of increasing soil moisture as the class moved from a1 to d3. In six of the reclaimed treatments (I, WA5, M, A, E, and FF), soil moisture was higher than the natural range; in the other treatments (F, H, Albion, and B), soil moisture was within the natural range of variability for the b3 to d3 ecotype mineral soils. Litter moisture was generally similar to FH moisture in all reclamation treatments, and FH moisture in the reclamation treatments was within the range of natural variability.

Soils in reclamation treatments were 1 to 2 pH units higher than natural ecotypes, except treatment FF which retained near-natural pH values (Fig. S2). On reclaimed sites (except FF) the pH(w) of litter ranged from 6.3 to 6.9, FH 6.3 to 7.2, and soil 6.1 to 7.5; in the natural ecotypes the pH(w) of litter ranged from 5 to 6.3, FH 4.9 to 5.8, and soil 4.4 to 5.5. The pH(w) of litter in natural ecotypes increased from pine (a1) to aspen and forbs with spruce (b1, b3, d1) and fell again as spruce and *Sphagnum* dominated (d3).

The C/N ratios of FH on reclamation treatments were below the natural range of variability (Fig. S3). An abandoned peat dump WA5 had the lowest C/N ratio in litter, soil, and the second lowest in FH. The C/N ratios in reclamation treatments generally followed the order: litter > soil > FH, which differs from the pattern in natural ecotypes where the C/N ratios decreased with depth (litter > FH > soil). The C/N ratio of natural soils decreased as site moisture status increased and the C/N ratios for all three d-ecotypes was low (<20).

Litter Decomposition

Mass loss of aspen litter on the reclamation treatments except A, I, and WA5 was below the range of natural variability (Fig. 2). Treatment B (plot 3) had the lowest mass loss (11.5%). Using SAS 9.1 a model was sought to explain decomposition rate of aspen litter across all treatments in Year 1 using variables of moisture, pH(w), and C/N ratios of both FH and soil. The only variable of significance ($P = 0.0021$) was soil C/N ratio (Fig. S4). The model ($r^2 = 0.5608$) is:

$$\% \text{mass loss} = 48 - 0.80(\text{C/N}_{\text{soil}})$$

There appeared to be two clusters. Group (i) describes soils of lower (≤ 20) C/N ratio that are found on ecotypes d1, d2, and d3, and treatment WA5. Group (ii) comprises other reclamation treatments, and ecotypes a1, b1, and b3, with higher soil C/N ratios.

Development of the Surface Organic Layer

The dry mass of FH on all reclaimed treatments fell below the range of natural variability (Fig. 3). Across the natural ecotypes the FH dry mass sample range was 168 to 1479 g (equivalent to 4.4–38.4 kg/m²) for a1 to d3 sites. The FH dry mass on natural sites varied significantly and negatively with pH(w) ($P < 0.05$, $r^2 = 0.3881$).

The FH development on *reclaimed* sites was best predicted with a model using site age (transformed to $\log_{10}$) and FH dry mass (transformed to natural $\log \ln$) ($P < 0.0001$, $r^2 = 0.8282$).

$$\ln(\text{dryFH}+1) = -0.25 + 3.46 \times \log_{10}(\text{age})$$

pH(w) and total carbon content of FH were also significant ($P < 0.05$) but each variable contributed just 1.5% to the model ($r^2 = 0.8579$). The FH moisture, FH depth, and FH total N were not significant. Applying the model as a best guess (since 17% of the variability remains unexplained), the minimum time by which FH will be within the lower range of natural variability (i.e., 168 g) is 36 yr.

In situ Bioavailable Nutrients

Reclaimed sites generally had high NO_3^- but low NH_4^+ , P, and K. Sodium and Mn were much reduced, but Ca, Mg, Fe, S, B, and Al were all generally elevated on reclamation treatments (Table S3). Copper was generally below detectable limits across all ecotypes and treatments, but trace amounts were detectable on some reclamation treatments. Zinc was generally present at very low amounts across all plots but at slightly elevated levels on reclamation treatments. The levels of nutrients on most reclamation treatments was outside (above or below) the natural range of variability. Albion had extremely low P and K but very high Fe and S. Albion and WA5, both unfertilized, had higher NO_3^- than the fertilized reclamation treatments and the natural ecotypes. Repeatedly fertilized sites H and I, compared with the once-only fertilized treatments (A, B, E, F, M, FF), generally had slightly higher mean values or an extended range of values (as indicated by the standard deviation) of most nutrients.

Plant Community

There was a general trend for reclaimed sites to have more bare ground, grasses, and forbs but less cover of moss, lichen, shrubs, broadleaves, spruce, pine, or woody debris than natural sites. The only treatments that had any lichen cover were H and I. There was considerable variability within and among ecotypes and treatments for all 10 groundcover classes measured. There was also variability with age irrespective of treatment, with younger and older reclaimed sites having fewer groundcover classes than middle-age ones, and with an apparent “tailing off” of cumulative groundcover after a peak at the 21 to 25 yr age class (Fig. 4). Albion and WA5 were the least similar to the other treatments, with Albion having almost nil plant cover while WA5 was densely populated by shrubs and broadleaves; note that WA5 was the sole representative of the 26 to 30 yr age class, accounting for the high broadleaf and shrub values for that age class. Among the natural ecotypes, d1 appeared to be the most different, having the lowest mean values for moss and lichen but the highest mean values for forbs, shrubs, and broadleaves.

Inclusion of groundcover data into the model for FH development on reclamation treatments indicated that shrub cover was a contributing factor ($P < 0.0001$) contributing 10% to the model, whereas age and pH contributed about 50 and 20%, respectively. Woody debris and pine were also significant ($P < 0.01$) but each contributed just 1.5% to the model. The full model ($r^2 = 0.8469$) was:

$$\ln(\text{dryFH} + 1) = 0.15 + 0.03(\text{age}) + 0.40(\text{pH}_{(w)}) + 0.04(\text{shrubs}) + 0.03(\text{pine}) + 0.09(\text{woody debris})$$

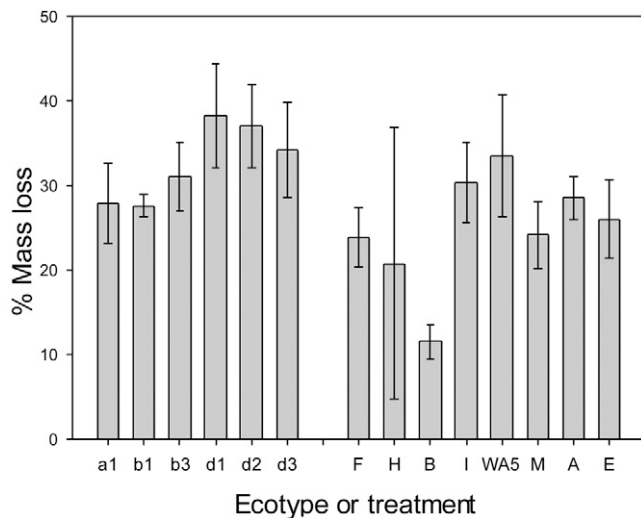


Fig. 2. Mean mass loss of aspen leaf litter during 1 yr of decomposition, (left) at natural ecotypes and (right) reclamation treatments, ranked in order of increasing soil moisture. Error bars indicate one standard deviation. The mean value for treatment H is the mean of three litter bags; all other treatment means are for seven litter bags.

Statistical Interpretation

Nonmetric Multidimensional Scaling

Nonmetric multidimensional scaling produced a solution in three dimensions with a final stress of 11 and a final instability of 0.00001 after 81 iterations. Most of the variability was expressed in axis 3 ($r^2 = 0.416$) which was correlated with mosses, lichen, grass, forbs, conifer, woody debris, pH(w), B, Ca, and Na (Table 2). The Monte Carlo test indicated that the proportion of 50 randomized runs with stress less than or equal to the observed stress (the probability that a similar final stress would be achieved by chance) was 0.0196, or 2%.

The ecotypes clustered together with some reclaimed sites from treatments E, H, and I (Fig. 5). Other reclamation treatments were more distant, so less similar to the ecotypes. In general, reclaimed sites by 21 to 25 yr, irrespective of treatment, were more similar to a natural forest and less similar to younger reclaimed site (Fig. 6). There were exceptions—the oldest plots for treatments A and B remained distant from the natural ecotypes after 24 and 19 yr, respectively. The eldest sites (age-class 31–35 yr) also appeared less similar to natural ecotypes and different from the other reclaimed plots, projecting forward in the figure.

Multiresponse Permutation Procedures

The test of chance-corrected within-group agreement, describing within-group homogeneity compared to random expectation (McCune and Grace, 2002) was 0.17. Treatments E and I as groups were indistinguishable from one another ($P = 0.71$) and not significantly different from the natural ecotypes group ($P > 0.05$). Treatment I, as a group, was not significantly different from ecotypes b1 and b3; treatment E, as a group, was not significantly different from ecotypes b1, d1 and d2 ($P > 0.05$). Treatment H (a tailings sand treatment) as a group differed from the natural ecotypes group ($P = 0.0004$), although several plots exhibited similar results to treatment I. This treatment was also

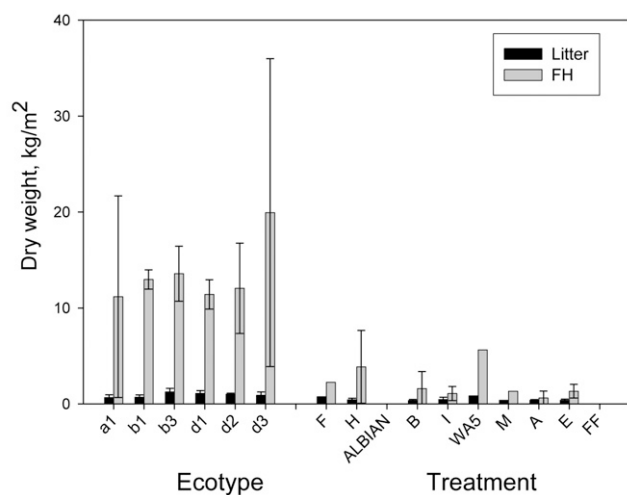


Fig. 3. Mass of surface organic matter per square meter at the start of the growing season at (left) natural and (right) reclaimed sites by treatment. Error bars represent one standard deviation.

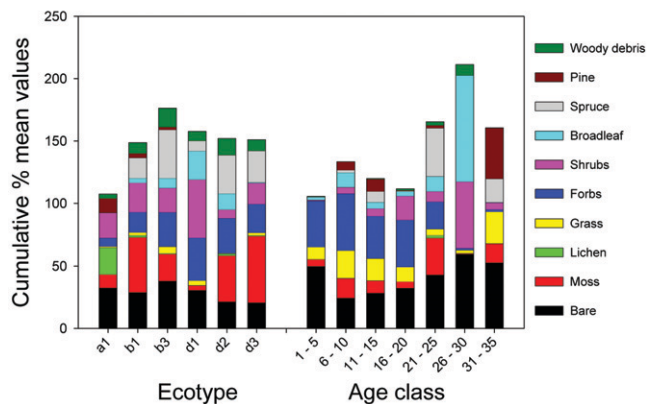


Fig. 4. Cumulative mean ground cover classes by ecotype or age class.

different from all other treatments ($P < 0.05$) except E and I. Treatments A and B (both tailings sand treatments) as groups were indistinguishable from each other ($P = 0.58$) and different from the natural ecotypes ($P < 0.00001$). Pairs of treatment groups that were significantly different from each other were A vs. H, B vs. H, A vs. I, and B vs. I ($P < 0.05$).

Cluster Analysis

Age-class data were used to identify temporal changes in relationships. The cluster dendrogram was pruned at the 50% level, revealing four distinct clusters, or groups (Table 3). Treatments E, H and I clustered more closely with natural ecotypes than other treatments.

Discussion

The reclamation treatments differed from natural forests and from each other in the physical and chemical characteristics measured. Broadly, the presence of tailings sands at depth led to reduced moisture content, while a P-M mix at the surface was associated with higher moisture unless the P-M mix was directly over tailings sands as in treatment H. Subsoil caps (treatments B, F) were generally dry.

Table 2. Pearson linear correlations (r) of variables vs. ordination axes indicating direction of relationship, positive or negative.

Variable	Ordination axis		
	1	2	3
Bare ground		-0.853	
Moss			0.625
Lichen			0.601
Grass			-0.632
Forbs		0.556	-0.681
Broadleaves	0.719		
Conifer	-0.646		0.550
Debris			0.697
pH(w) soil			-0.535
NO ₃ -N		-0.494	
Ca			-0.610
Boron			-0.522
Na			0.542

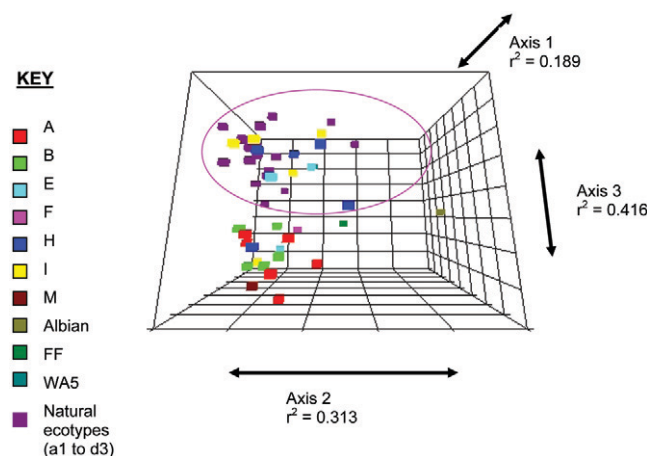


Fig. 5. Nonmetric multidimensional scaling plot in three dimensions, by treatment or grouped natural ecotypes. This three-dimensional image should be viewed like a hollow cube or a box, looking down into it from above with its top surface (the widest part of the image) in the same plane as this page. Each plot is represented by a single dot, some are partly hidden. The plots are suspended in space with those closer to the top appearing larger than those found farther along axis 1 (into the page). The pink overlay contains those plots within the natural cluster.

The high pH of the P-M mixtures at most reclamation treatments when compared to the natural soils can be attributed to the inherently high pH of salvaged mineral materials of the Athabasca oil sand region (Fung and Macyk, 2000), and mixing this alkaline material with surface organic material during overstripping. Windblown deposits from exposed tailings may also have contributed. Treatment FF was the only treatment with a pH similar to the natural ecotypes. This P-M mix had an unusually high C content (191 g OC kg⁻¹) and was comprised of undecomposed, naturally acidic fibric peat (Hemstock, 2008).

The reclamation treatments differed from natural ecotypes in their nutrient status. In all treatments, fertilized or unfertilized, N was present at levels at or above the natural range. Addition of N in fertilizers should be revisited in light of these observations of abundant nitrate on unfertilized sites. Those treatments with levels of both K and P close to the natural ranges were: (i) those with a shallow P-M cap that received fertilizer for several years (treat-

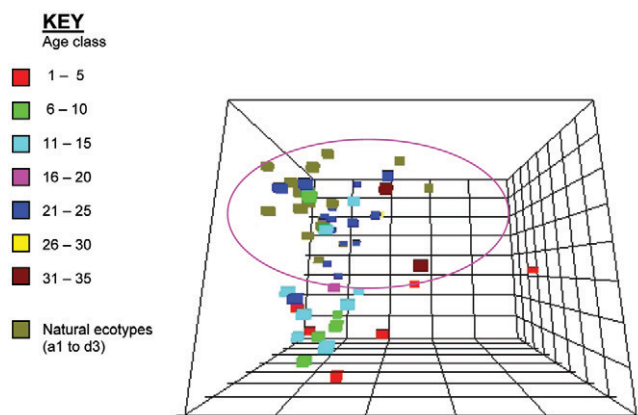


Fig. 6. Nonmetric multidimensional scaling plot in three dimensions, by age class or grouped natural ecotypes.

ments H and I), (ii) those with a shallow P-M cap over nonsaline, clay-rich subsoil and overburden, fertilized once (treatment E), and (iii) those with a deep mineral-soil cap laid over tailings sands and fertilized once (treatment B). This includes both young and more mature plots, indicating that it is a treatment effect rather than an age effect. Other P-M treatments were characterized by very low K and/or P levels, which may indicate that these nutrients, rather than N, are deficient at these sites.

Development of a surface organic layer varied among the treatments although the differences between treatments were not large. The model suggested that it would take at least 36 yr to achieve an FH layer similar to that found in an a1 ecotype, this being at the lowest end of the natural range of variability. In the field, only plot 30 (treatment H) had a dry mass FH close to the lower end of the natural range; this plot was 34 yr old. Interestingly, Frouz et al., (2001) found that the development of forest floor (L, F, and H layers) increased for the first 30 yr after reclamation of two open-cast coal mines in eastern Europe, after which they appeared to stabilize.

The model indicated that shrubs were important in the development of FH, so planting shrubs is recommended to accelerate forest-floor development. Castro et al. (2004) also noted that shrub cover aided reforestation in a harsh environment by mitigating water stress through reduced solar radiation to the floor, and also helped prevent seedling mortality from browsing. It has also been shown that the leeward sides of shrubs benefit soil NH₄⁺ pools and soil organic matter, presumably by interfering with wind and thereby creating localized changes in microclimate and plant litter deposition (Mummey et al., 2002). Shrub cover may play a similar role in reclaimed oil sands sites with an additional benefit of facilitating FH development.

Litter decomposition appeared to be depressed on the reclamation treatments compared with that on natural ecotypes despite moisture levels generally above the natural range of variability. Decomposition rate was especially low on treatment B, a subsoil capping mixture laid over tailings sands. In treatment F, where subsoil is laid as a cap over overburden, although it was drier than treatment B, decomposition was faster but still below than that of most P-M treatments. This suggests that if tailings sand, which has a higher hydrocarbon content than

Table 3. Groups of plots as defined by cluster analysis, by ecotype and treatment by age class.

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overburden, is reclaimed with a subsoil cap, the tailings sand may reduce that subsoil's inherent microbial activity. Oil addition to soil results in high C/N ratios which is unfavorable to microbial activity, though this can be overcome by nutrient addition (Choi et al. 2002). This hypothesis is being tested in a parallel study of soil microbial diversity and function at these same sites (P. Dimitriu, personal communication 2007). Tailings sand, in contrast to overburden and peat, has been shown to be devoid of ectomycorrhizae, but mycorrhizae can be restored using peat additions (Bois et al. 2005).

Plant diversity on reclamation treatments appeared to remain fairly stable before declining with advancing canopy closure at about age 31 to 35, when understorey plants disappeared. This is analogous to the decline in ruderal species noted by Frouz et al., (2008) as trees came to dominate postmining sites in the Czech Republic, about 30 yr into succession. There was some evidence that grasses form a higher proportion of plant cover on our reclaimed sites than on the natural sites; this has also been reported on highly disturbed reclaimed sites where the plant species originally sown were still evident after 14 (Norman et al., 2006), 25 (Rayfield et al., 2005), and 45 yr (Hodačová and Prach, 2003).

Some of the reclaimed sites appear to be on a path to becoming functioning ecosystems similar to those naturally present in the boreal region; others do not. The best-performing P-M treatments are E, H, and I. There was a line of separation at the 16 to 20 yr age range where reclaimed sites E, H, and I shifted away from their reclaimed site status and toward the natural range. These treatments also showed high diversity with 8 to 10 different groundcover types. Treatment E appeared to become similar to natural ecotypes within 15 to 20 yr, and Treatments H and I within 20 to 25 yr.

The worst-performing treatments, A and B (both covering tailings sand), appear to remain in the early reclamation phase for at least 25 yr. These treatments appear to be on a trajectory to a novel ecosystem that may or may not achieve the objectives of restoring commercial forestry and wildlife habitat potential within an appropriate timescale. It may be possible to divert these treatments to a desired trajectory by adding more fertilizer in the early years as per treatments H and I.

Since there was only one replicate in each of treatments F, M, FF, WA5, and Albian, we are less confident in drawing conclusions as to their candidacy but it looks very likely that Albian,

an outlier separate from all other treatments or natural ecotypes, will take decades to shift out of its reclamation state. Treatment WA5 was clustering with the natural ecotypes, but it is not feasible on a large scale since it is a deep peat treatment and the amount of available peat is limited. As for treatments M (age 6) and FF (age 4), which differ in the type of peat used (mesic vs. fibric), there is little evidence yet that either of these young sites is moving toward a natural condition. They should continue to be monitored to determine relative advantages of the different peat types, and fertilized to overcome low availability of K and (on treatment M) Mn. Treatment F (age 14) may be of particular interest when peat resources decline since it has no surface peat amendment and, with the exceptions of K and Mn, it is broadly similar to the range of variability in plant nutrient chemistry of natural ecotypes. Shrubs could be planted on this treatment to enhance its diversity and boost its restoration trajectory.

The approach used—comparing two ecological attributes (plant community and ecological processes) on young reclaimed sites with those on a range of older, natural forest ecotypes—appears to have been successful in allowing us to determine which treatments are creating functional ecosystems similar to the target ecotypes. The comparison would have benefitted from inclusion of natural sites that had been disturbed (burned, blown down, or logged) within the last 20 yr. Studying natural ecosystem recovery would allow us to see whether ecosystem development on reclaimed sites mirrors that on natural sites of the same age since disturbance. This comparison was originally planned, but was not possible with the available resources. Additional information on microbial diversity, organic matter quality, and hydrological function is being collected at the sites used in this study, which will improve our ability to measure and predict the success of reclamation treatments in the oil sands.

Nonmetric multidimensional scaling proved to be a useful technique for analyzing data from this unbalanced experimental design—it rapidly produces results that are visually informative and can be quickly interpreted. Reclamation treatments were still distinguishable after 20 to 25 yr and treatment differences created consistent, observable differences in response variables. Results of the cluster analysis confirmed those of NMS.

It was interesting that some of the older reclaimed sites which looked like forests on casual observation, and might have been assumed to have been successfully restored, were in fact not on a trajectory to becoming similar to one of the target ecotypes.

Instead, they appear to be developing into a new nontarget ecotype. Does this mean that restoration of these sites has been unsuccessful? It may be that expectation of restoring an ecosystem precisely like that which was perturbed is unrealistic. Perhaps when dealing with a profoundly-disturbed landscape on this scale, recreating an ecological landscape (i.e., a mosaic of different communities comprising forest, forest edge, grassland) of functioning ecosystems is a more realistic and sufficient goal.

Conclusions

Some of the reclamation treatments applied to oil sands sites appear to develop into ecosystems with similar plant community composition and soil nutrient availability to those which occur naturally in this area. Other reclamation treatments did not closely resemble the natural forest ecotypes and may become novel ecosystems. It appears that treatments that use: (i) a repeatedly-fertilized peat-mineral mixture, or (ii) a peat-mineral mixture fertilized once and with underlying subsoil over clean overburden, are developing into a functioning forest soil capable of supporting ecosystem processes which mimic those of the natural boreal forest.

Implications for Practice

Sites reclaimed with a peat-mineral cap require fertilizer with P, K, and trace elements including Mn to provide an early boost to ecosystem development. Annual fertilizer treatments for the first 5 yr after planting may be sufficient to boost a peat-mineral-reclaimed site to a status similar to a natural forest by age 20 to 25, whether over tailings sands or not. Repeated addition of fertilizer will assist plant growth and litter production, and hence development of a surface FH layer. Lesser applications of fertilizer may be appropriate on nontailings sand sites capped with subsoil, with or without a peat-mineral layer at the surface, and with shrubs included in the planting mixture on those sites without a peat-mineral surface layer. The present reserves of relatively clean and unaltered peat and glacial till resources are unlikely to provide sufficient capping material for future reclamation on the scale currently proposed, so alternative prescriptions requiring smaller amounts and/or alternative inputs are needed. The practices of adding N in fertilizer should be revisited in light of the high levels of soil N measured in the reclamation treatments.

Supplemental Information Available

Table S1 describes the materials used for reclamation on oil sands sites. Table S2 describes the ecotypes used in this study. Figure S1 shows the moisture content of soils at each of the sites. Figure S2 shows the pH of soils at each of the sites. Figure S3 shows the V/N ratios of soils at each of the sites. Figure S4 shows the relationship between soil C/N ratio and mass loss of litter. Table S3 shows the bioavailability of nutrients in soils at each of the sites. This material is available free of charge at <http://jeq.scijournals.org>.

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